```c
// 1

#include <stdio.h>
#include <string.h>

char stack[100];
int top = -1;

int main() {
  char kelime[100];
  int i, n, palindrom = 1;

  printf("Kelime: ");
  scanf("%s", kelime);
  n = strlen(kelime);

  for (i = 0; i < n; i++) {
    top++;
    stack[top] = kelime[i];
  }

  for (i = 0; i < n; i++) {
    if (stack[top] != kelime[i])
      palindrom = 0;
    top--;
  }

  if (palindrom == 1)
    printf("Palindrom\n");
  else
    printf("Palindrom degil\n");

  return 0;
}

// 2

#include <stdio.h>
#define boyut 10

struct stack {
  int top;
  int deger[boyut];
};

struct stack s;

void push(struct stack *ps, int x) {
  if (ps->top == boyut)
    printf("stack dolu\n");
  else {
```

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    ps->deger[ps->top] = x;
    ps->top++;
  }
}

void pop(struct stack *ps) {
  if (ps->top == 0)
    printf("stack bos\n");
  else
    ps->top--;
}

void listele(struct stack *ps) {
  int a;
  if (ps->top == 0)
    printf("Liste bos\n");
  else {
    printf("Listeniz: ");
    for (a = 0; a < ps->top; a++)
      printf("%d ", ps->deger[a]);
    printf("\n");
  }
}

int main() {
  int secim = 0, sayi;

  while (secim != 6) {
    printf("\n1 Ekle 2 Sil 3 Listele 4 Tepeyi goster 5 Eleman sayisi 6 Cikis\n");
    scanf("%d", &secim);

    if (secim == 1) {
      printf("Sayi: ");
      scanf("%d", &sayi);
      push(&s, sayi);
    }
    else if (secim == 2)
      pop(&s);
    else if (secim == 3)
      listele(&s);
    else if (secim == 4) {
      if (s.top == 0)
        printf("Yigin bos\n");
      else
        printf("Tepe: %d\n", s.deger[s.top - 1]);
    }
    else if (secim == 5)
      printf("Eleman sayisi: %d\n", s.top);
  }
  return 0;
}
```

```c
// 3

#include <stdio.h>
#include <stdlib.h>

typedef struct node {
  void* dataPtr;
  struct node* link;
} STACK_NODE;

typedef struct {
  int count;
  STACK_NODE* top;
} STACK;

STACK* createStack() {
  STACK* stack = (STACK*)malloc(sizeof(STACK));
  stack->count = 0;
  stack->top = NULL;
  return stack;
}

void pushStack(STACK* stack, void* dataInPtr) {
  STACK_NODE* newPtr = (STACK_NODE*)malloc(sizeof(STACK_NODE));
  newPtr->dataPtr = dataInPtr;
  newPtr->link = stack->top;
  stack->top = newPtr;
  stack->count++;
}

void* popStack(STACK* stack) {
  STACK_NODE* temp = stack->top;
  void* data = temp->dataPtr;
  stack->top = temp->link;
  stack->count--;
  free(temp);
  return data;
}

int main() {
  STACK* stack = createStack();
  int *p;
  int i;
  int sayilar[3] = {5, 15, 25};

  for (i = 0; i < 3; i++) {
    p = (int*)malloc(sizeof(int));
    *p = sayilar[i];
    pushStack(stack, p);
  }
```

```
  while (stack->count > 0) {
    p = (int*)popStack(stack);
    printf("%d\n", *p);
    free(p);
  }

  free(stack);
  return 0;
}
```